

The Diderot Effect in Beauty Consumption

Language-First NLP Validation | 14,984 Amazon Reviews | 1,620 Users | Aug 2004 - Aug 2023

H2: Spending Spiral

-49%

Purchase Interval

76 days -> 39 days

H1: Linguistic Shift

+36.7%

Congruence Language

0.149 -> 0.200 hits/review

Cox PH: Phase Effect

HR = 1.24

Purchase Hazard

Congruence phase | $p < 0.0001$

Cox PH: Embeddings Win

HR = 1.07

Embedding Predictor

Lagged aesthetic score | $p = 0.0001$

Change Point Detection

18.8%

Regime Shifts Detected

200 of 1,062 eligible users

Kaplan-Meier

$p = 6.4e-12$

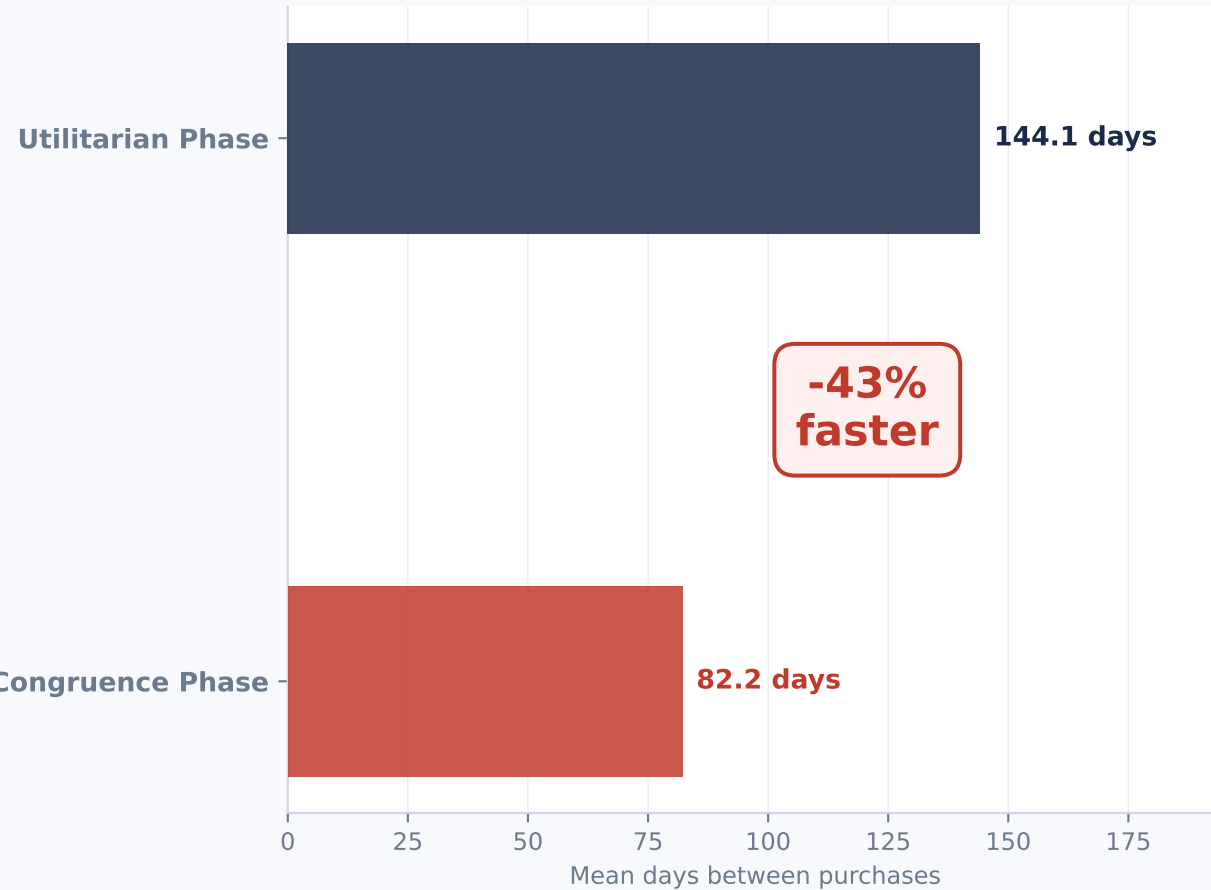
Log-Rank Test

Survival curves definitively differ

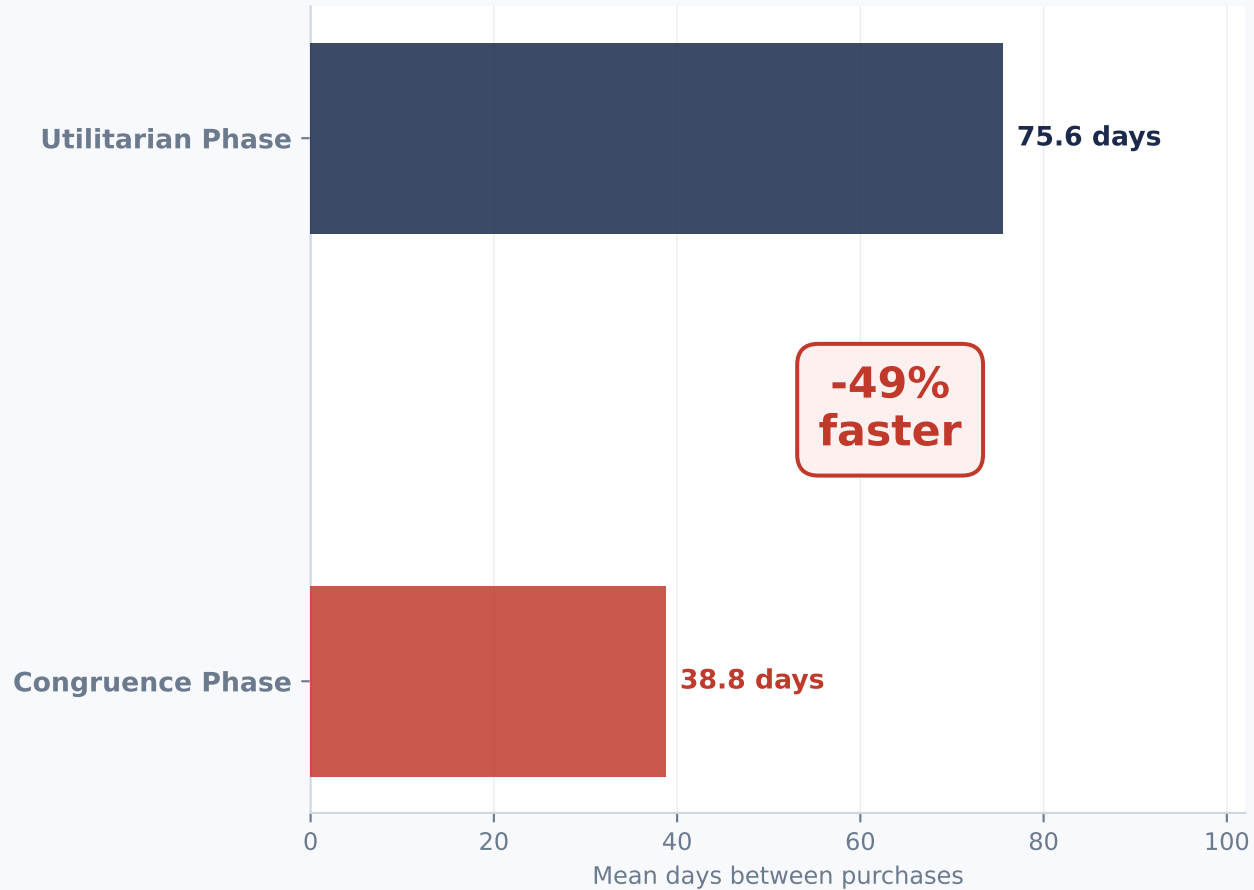
H2 Validation: Purchase Acceleration After the Aesthetic Trigger

Mean inter-purchase interval (days) falls significantly in the Congruence phase

Threshold Method ($S > 0.6$)

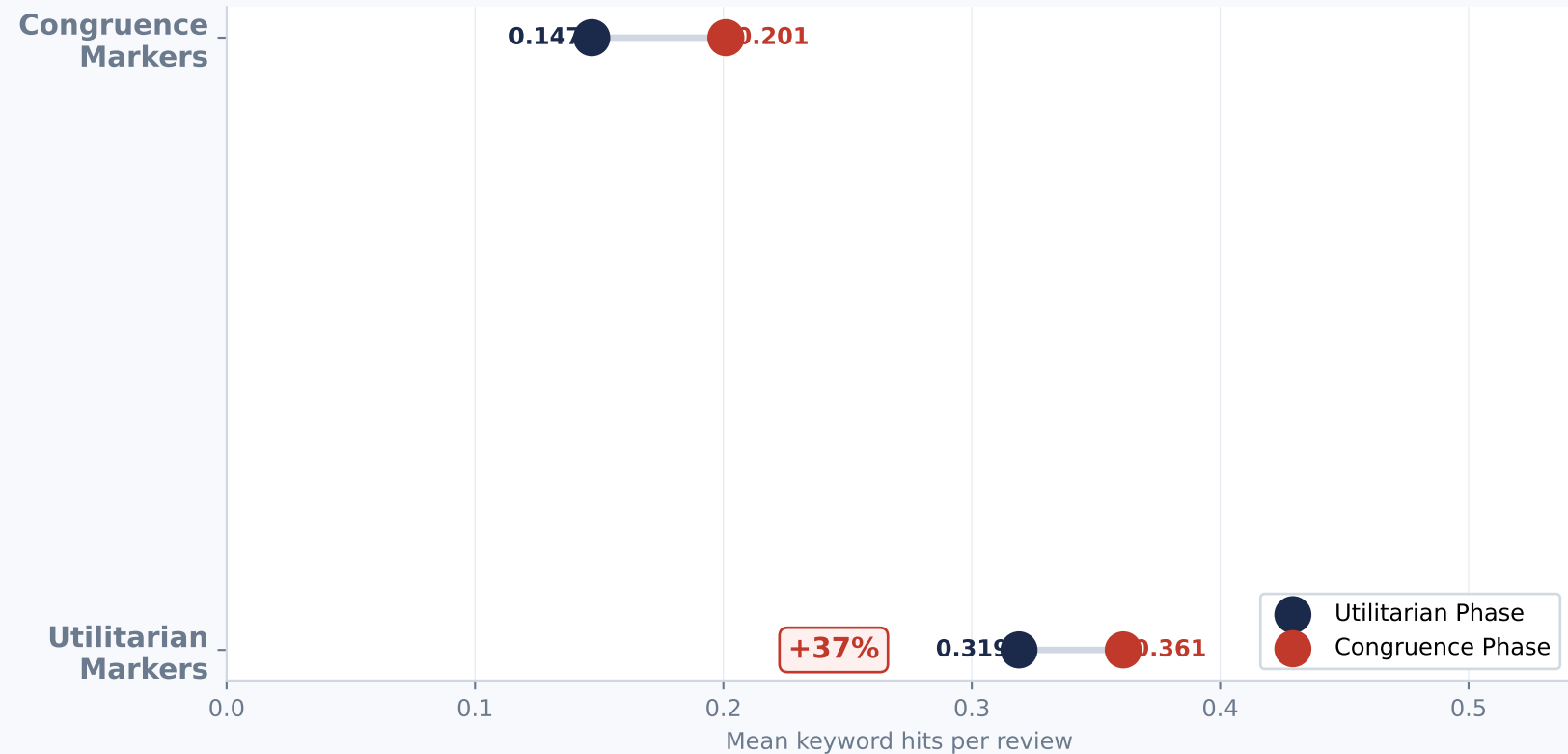


Data-Driven Method (ruptures Pelt)



H1 Validation: Linguistic Shift from Utilitarian to Congruence Language

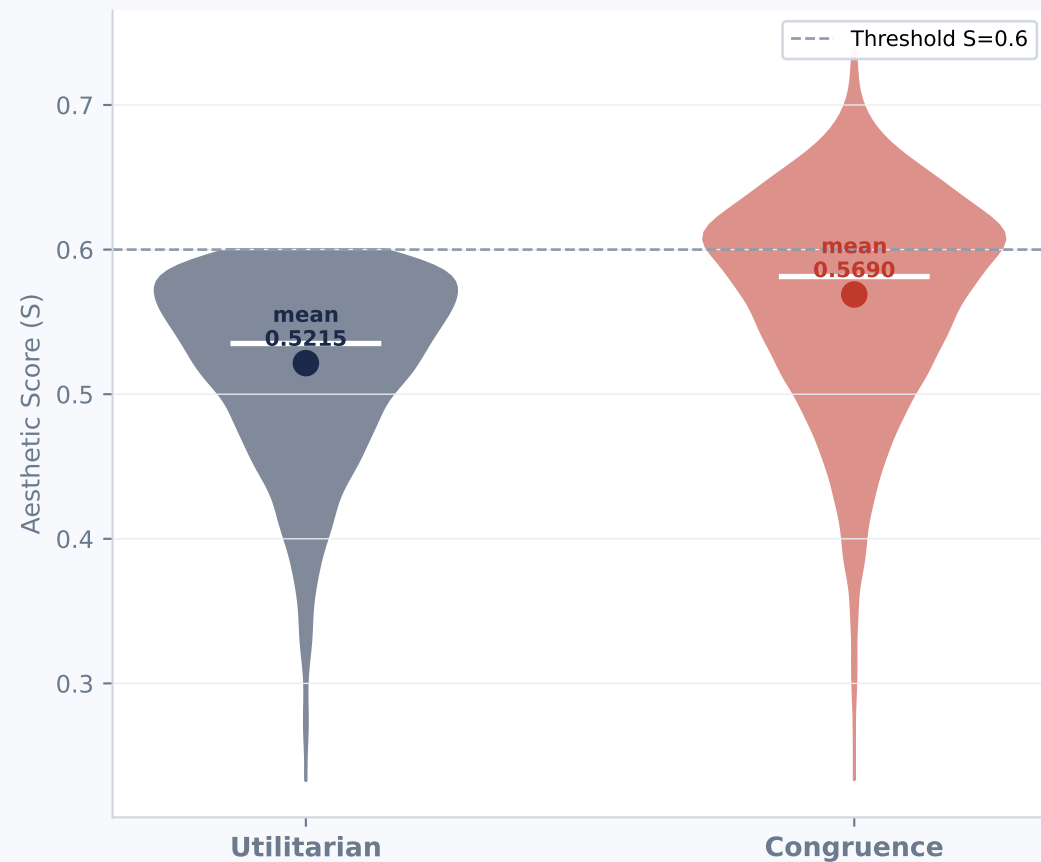
Congruence markers increase +36.7% post-trigger; utilitarian markers remain stable



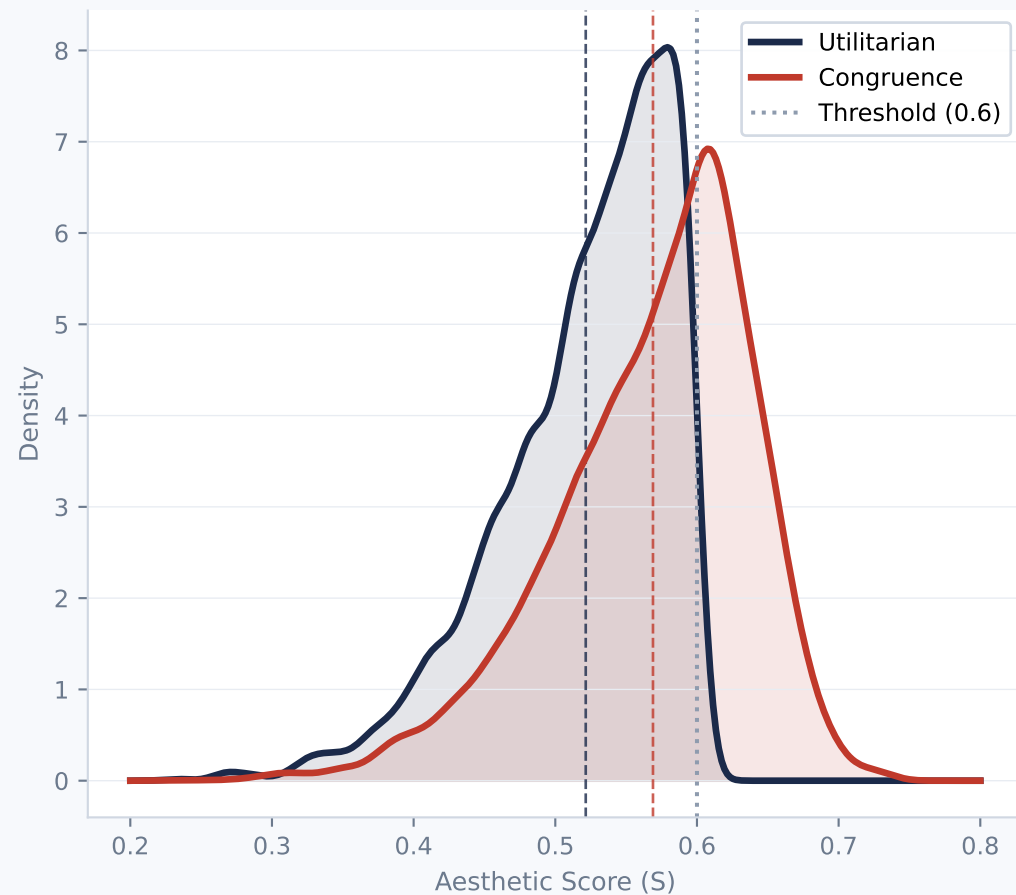
Embedding Validation: Aesthetic Score Shifts After the Trigger

Transformer cosine similarity to the aesthetic anchor rises in the Congruence phase

Score Distribution by Phase



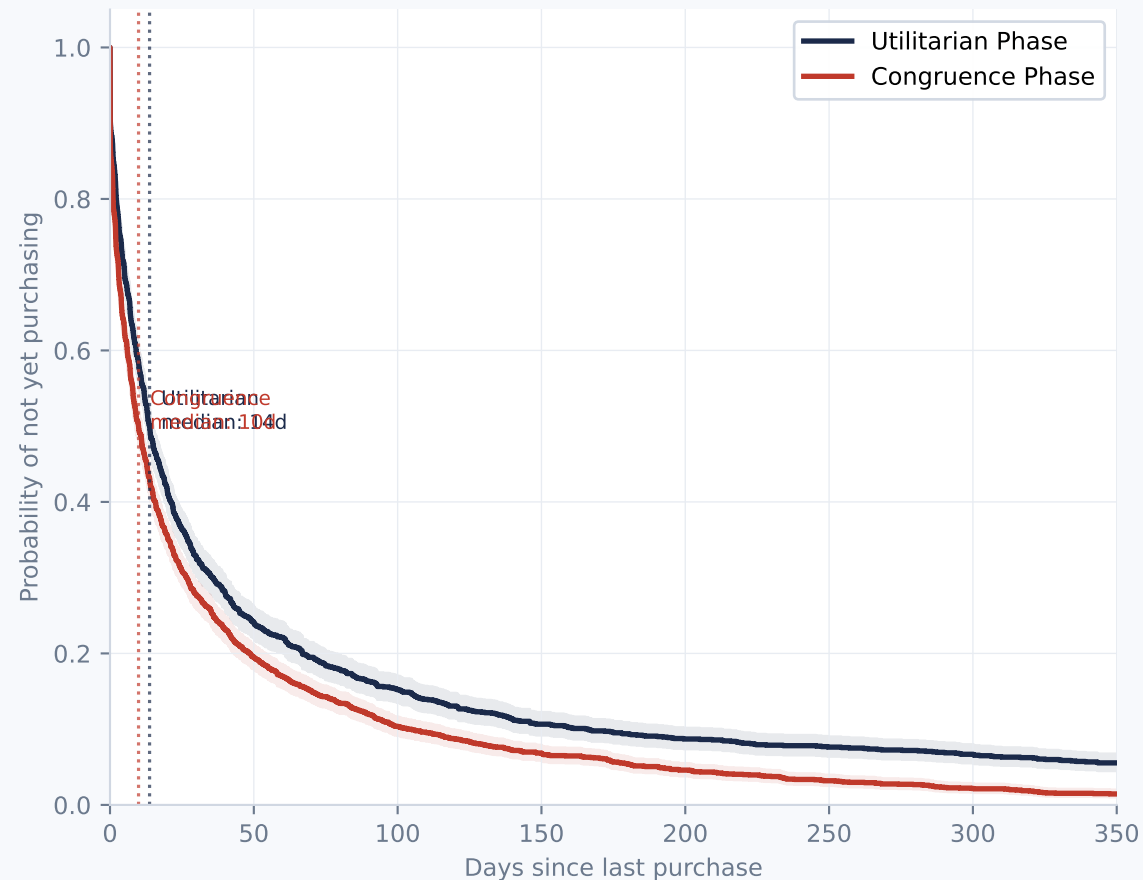
Score Density — Congruence Phase Shifts Right



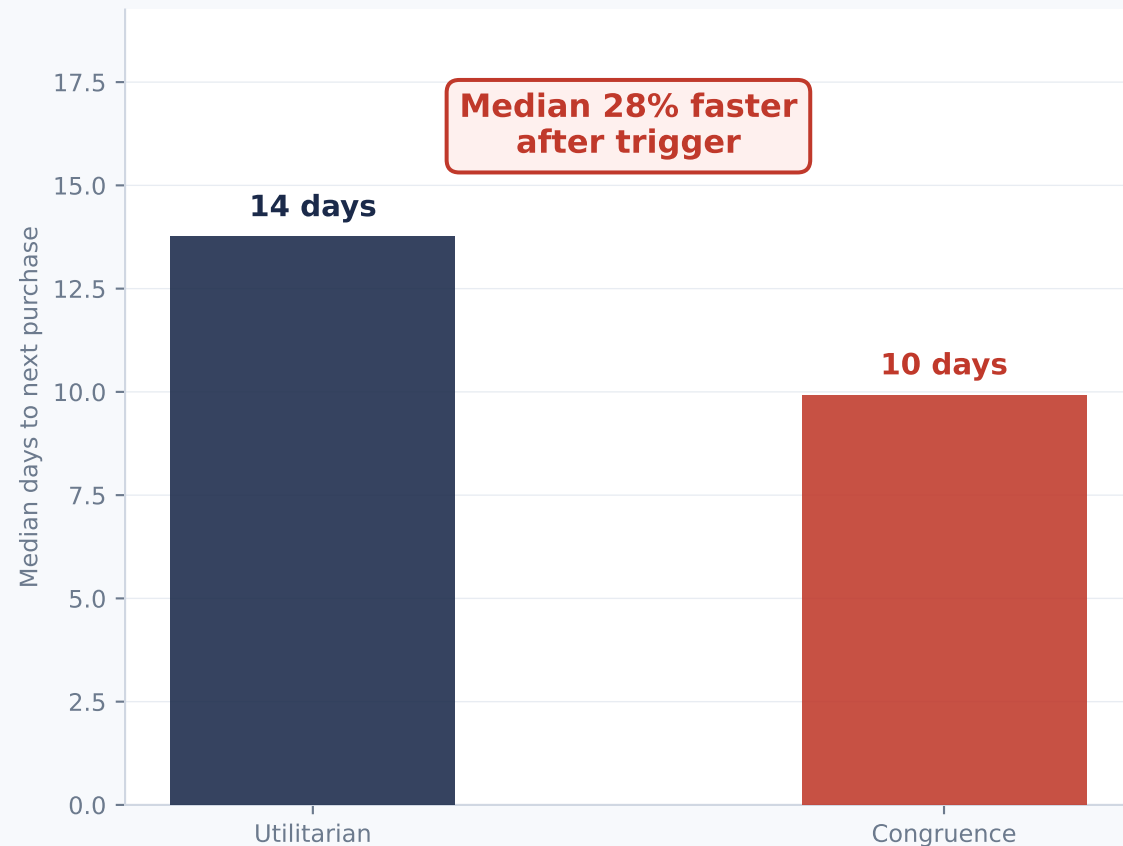
Survival Analysis: Congruence Phase Users Purchase Significantly Faster

Log-rank $p = 6.42\text{e-}12$ | Survival curves are definitively different between phases

Kaplan-Meier Survival Functions

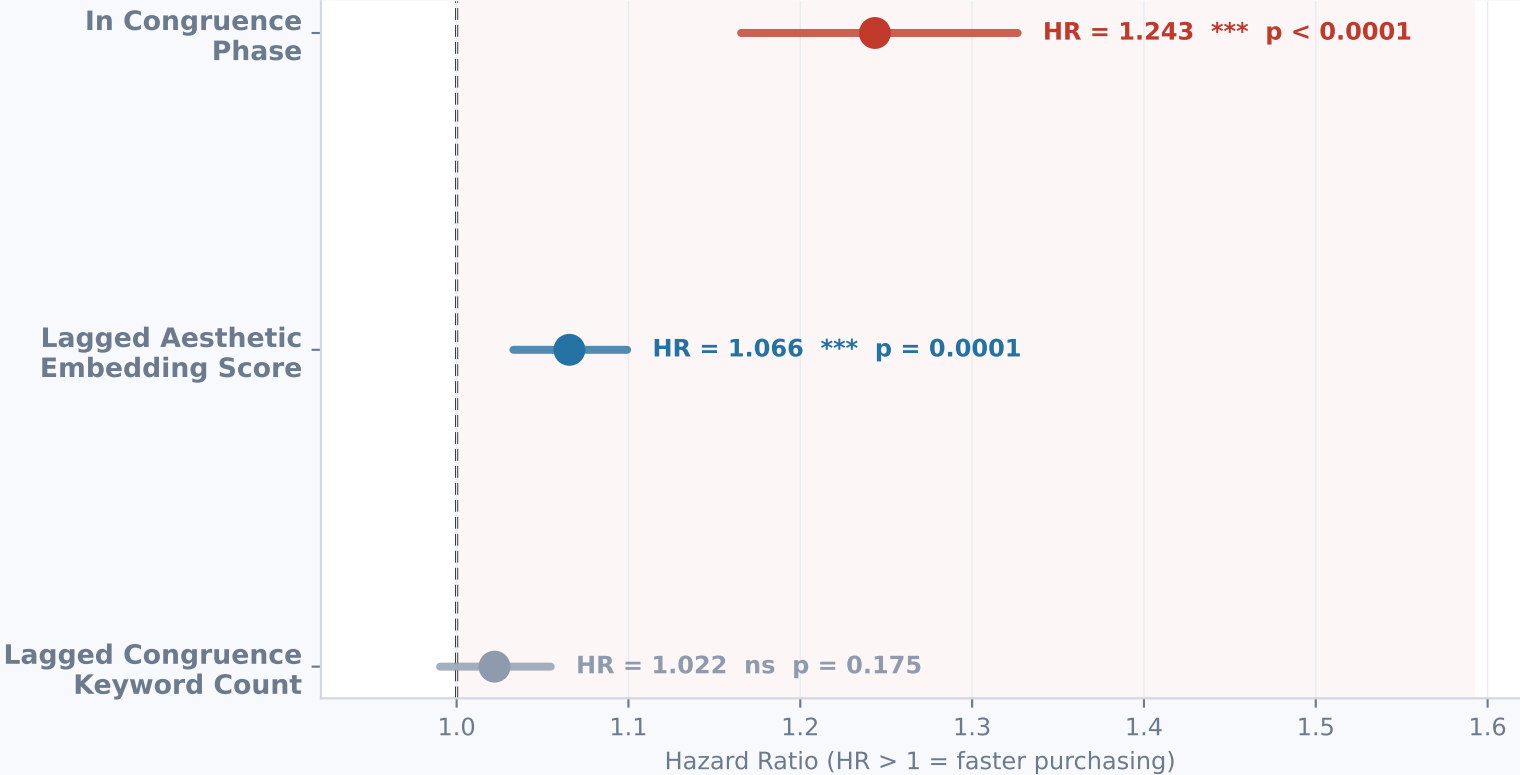


Median Survival Time by Phase



Cox PH Model: Transformer Embeddings Predict Purchase Acceleration; Keywords Do Not

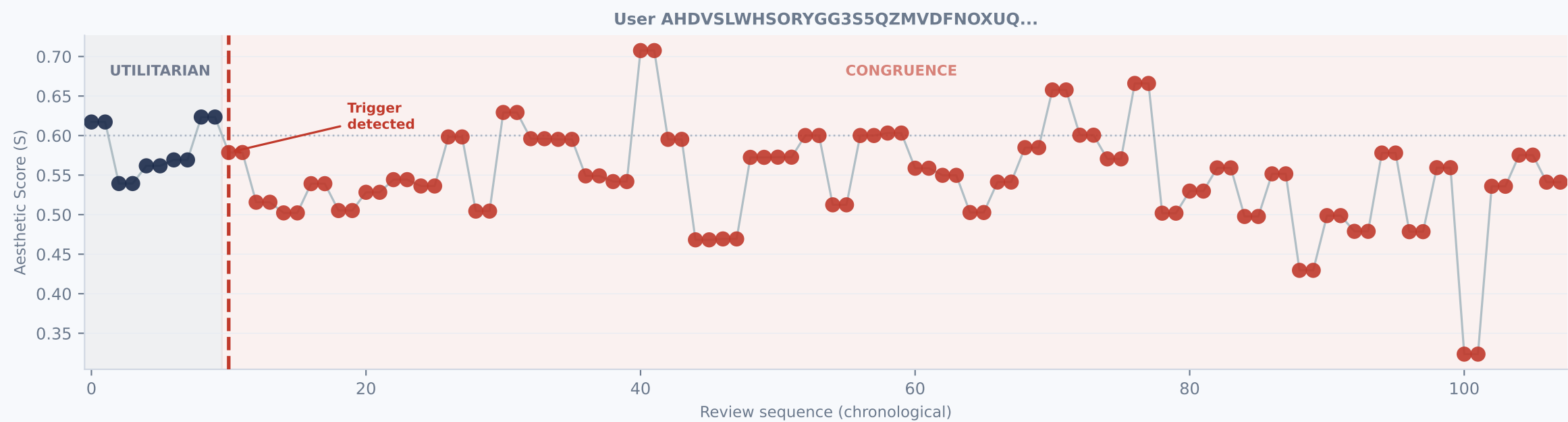
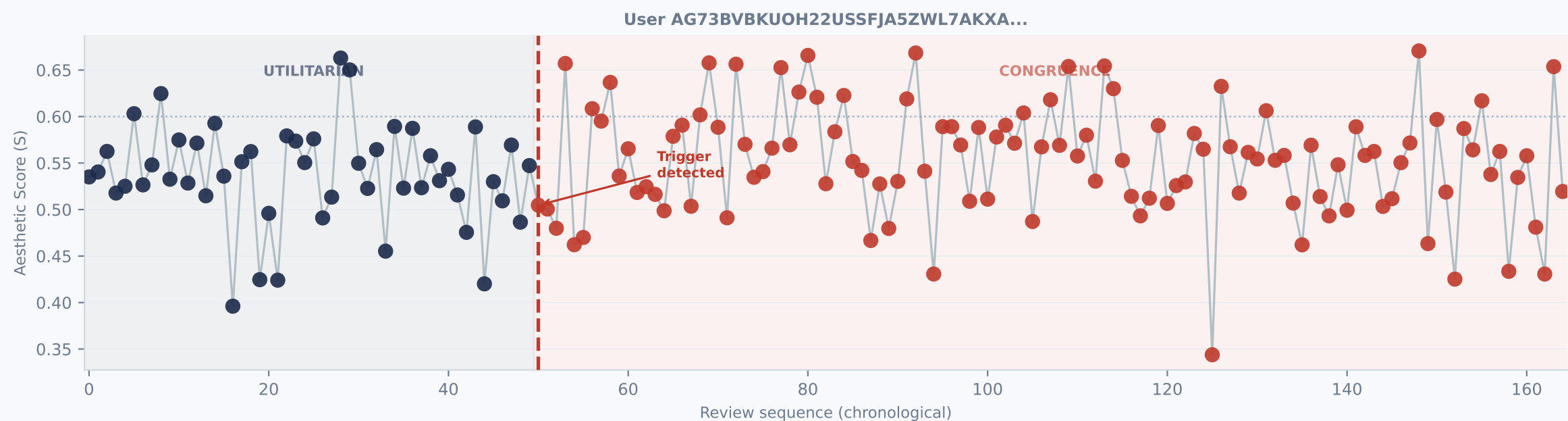
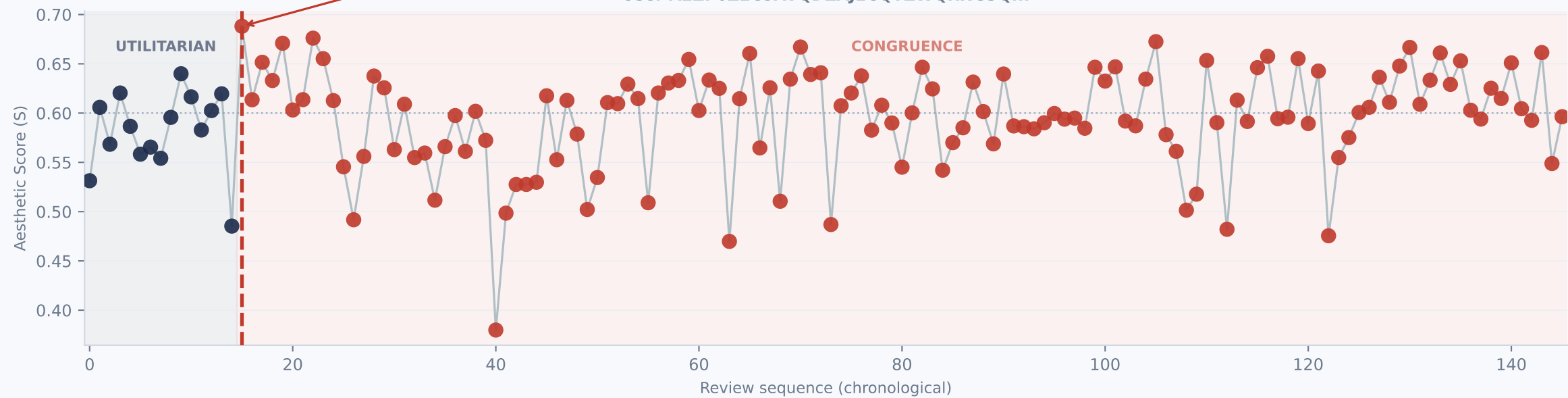
Hazard Ratio > 1 means higher purchase rate | Covariates standardised | n = 3,937



Key finding: Embedding score (HR=1.07, p=0.0001) predicts purchase acceleration. Keyword count (HR=1.02, p=0.175) does not — semantic representations outperform lexical ones.

Individual User Journeys: The Diderot Spiral in Action

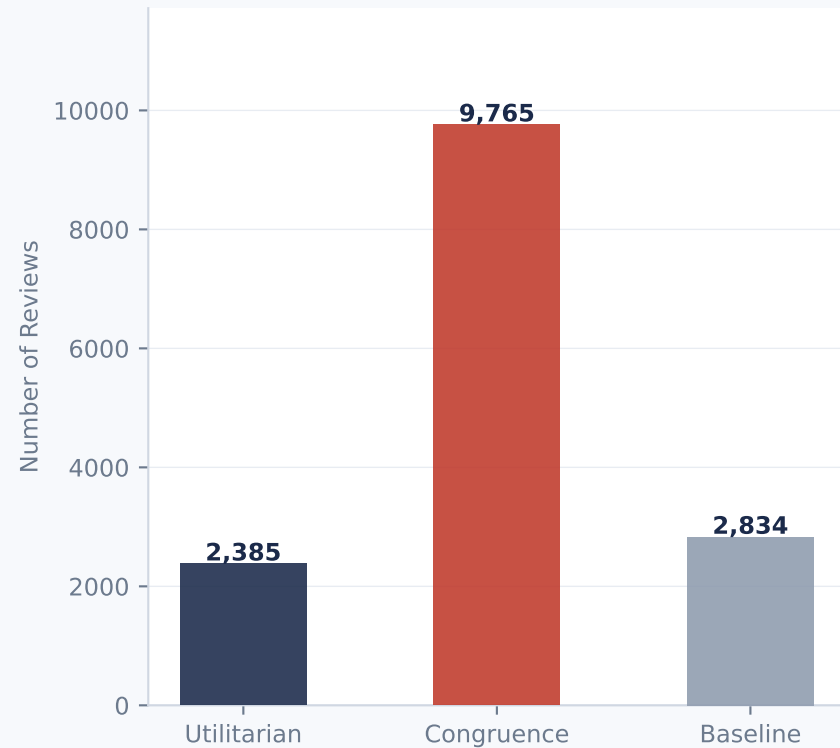
Each panel shows one user's aesthetic score over time — the red dashed line marks the detected regime shift



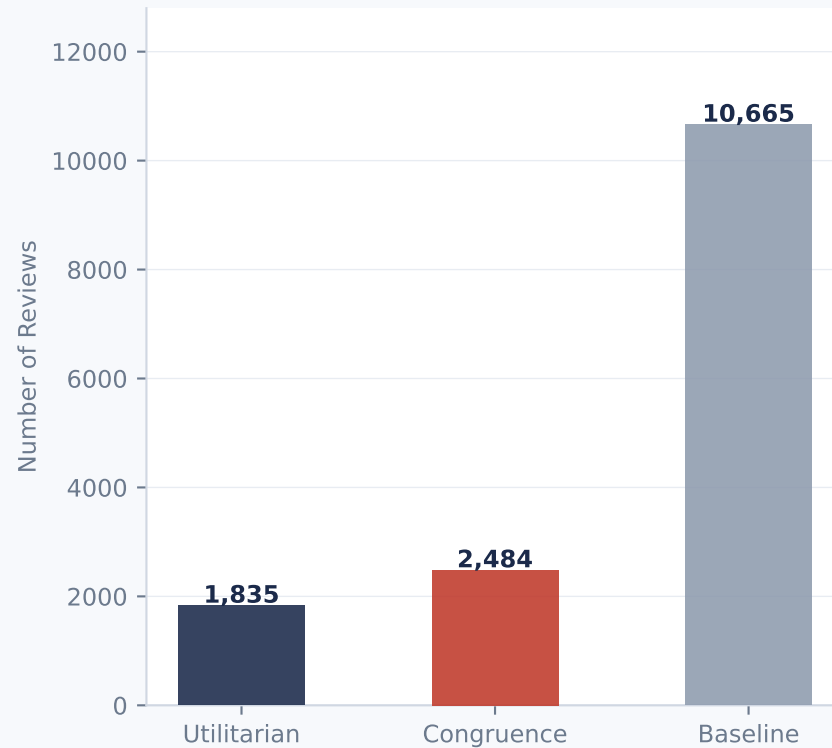
Threshold vs. Data-Driven Phase Detection: Both Methods Confirm the Spiral

The ruptures Pelt algorithm finds regime shifts without any hardcoded threshold

Threshold Method
($S > 0.6$)



Data-Driven Method
(ruptures Pelt)



Purchase Velocity: Both Methods

